

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

COMPUTER GRAPHICS

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SCOPE

2D TRANSLATION

2D translation Overview

2D transformation types

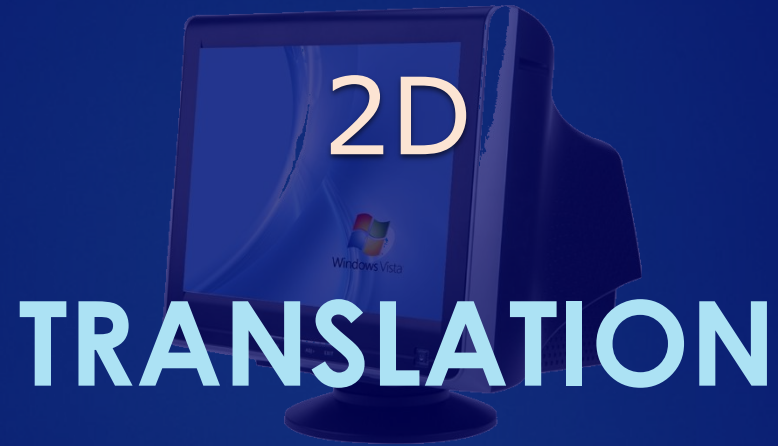
2D modelling transformation overview

Basics of geometric transformations

Example of 2D translation

2D translation algorithm

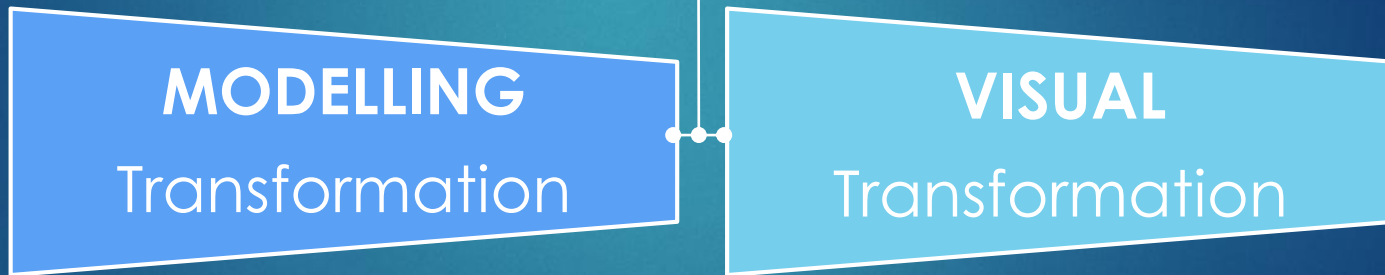




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2D TRANSLATION OVERVIEW?

2D translation or transformation is to **reposition** or **change** the **position** of the **objects(Polygonals)** in **two dimensional space** in computer graphics, in 2D transformation **animations** are produced by **moving** objects in the scene.



MODELLING TRANSFORMATIONS?

Is a geometric transformations where the coordinates of the objects are altered to change the geometry of the objects.

➤ BASICS OF GEOMETRIC TRANSFORMATIONS

Translation

Rotation

Scaling

Reflection

Shear



TRANSLATION EXPLAINED

A translation is applied to an object by repositioning it along a straight-line path from one coordinate location to another.

We translate a two-dimensional point by adding translation distances, t_x and t_y to the original coordinate position (x,y) to move the point to a new position (x',y')

The translation pair (t_x, t_y) is called translation vector or shift vector. We can express the translation expression in a matrix form by representing them as columns as follows:

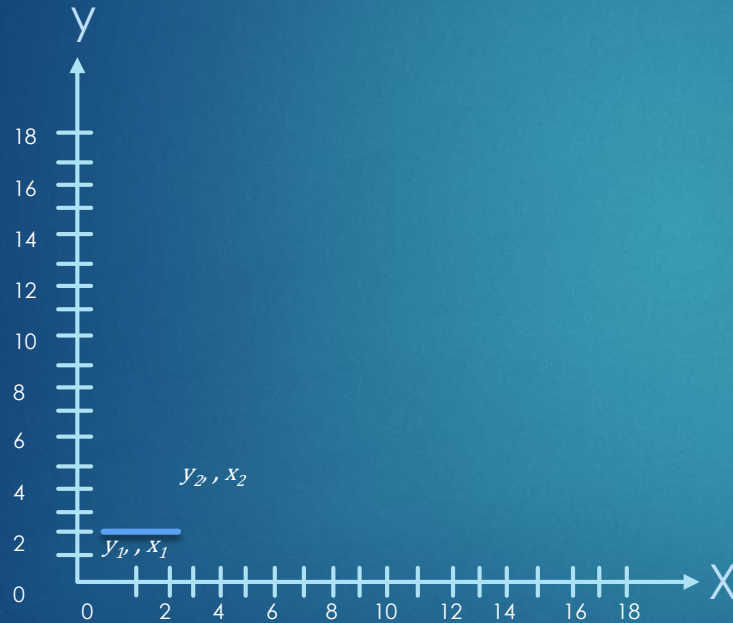
$$P = \begin{bmatrix} X_1 \\ Y_1 \end{bmatrix} \quad P' = \begin{bmatrix} X_{1'} \\ Y_{1'} \end{bmatrix} \quad T = \begin{bmatrix} t_x \\ t_y \end{bmatrix} \quad P' = P + T$$

$$X_1 = X + t_x$$
$$Y_1 = Y + t_y$$



EXAMPLE OF TRANSLATION

- ✓ Reposition a line with these coordinates $X1(2,2)$ and $Y1(2,1)$ into new position of $t_x(2)$ and $t_y(2)$.



$$P' = P + T$$

$$\begin{aligned} X_1' &= X_1 + t_x \\ X_1' &= 2 + 2 = 4 \end{aligned}$$

$$\begin{aligned} Y_1' &= Y_1 + t_y \\ Y_1' &= 2 + 2 = 4 \end{aligned}$$

$$X1 = (4, 4)$$

$$\begin{aligned} X_1' &= X_1 + t_x \\ X_1' &= 2 + 2 = 4 \end{aligned}$$

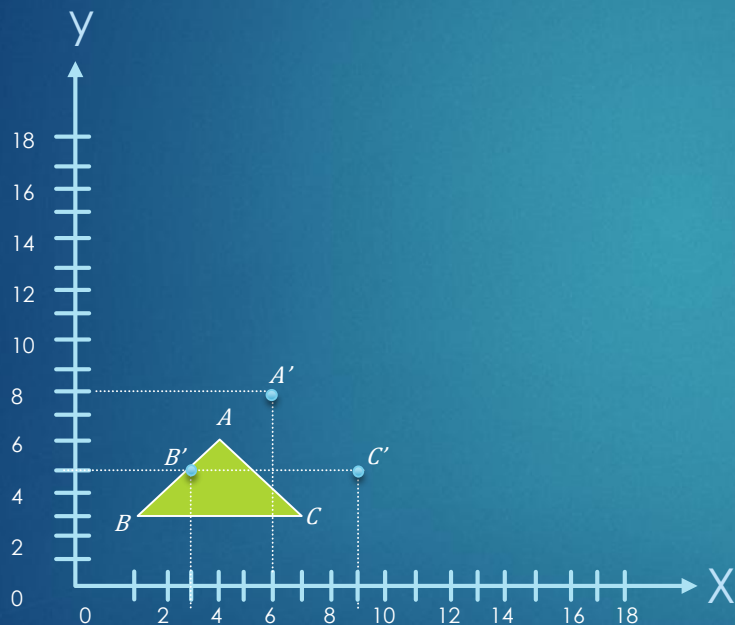
$$\begin{aligned} Y_1' &= Y_1 + t_y \\ Y_1' &= 1 + 2 = 3 \end{aligned}$$

$$Y1 = (4, 3)$$



EXAMPLE OF TRANSLATION

- ✓ Reposition the triangle drawn on the bellow graph into new position of $t_x = (2)$ and $t_y = (2)$.



$$P' = P + T$$

$$A = \begin{bmatrix} 4 \\ 6 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

$$A = (6, 8)$$

$$B = \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$B = (4, 6)$$

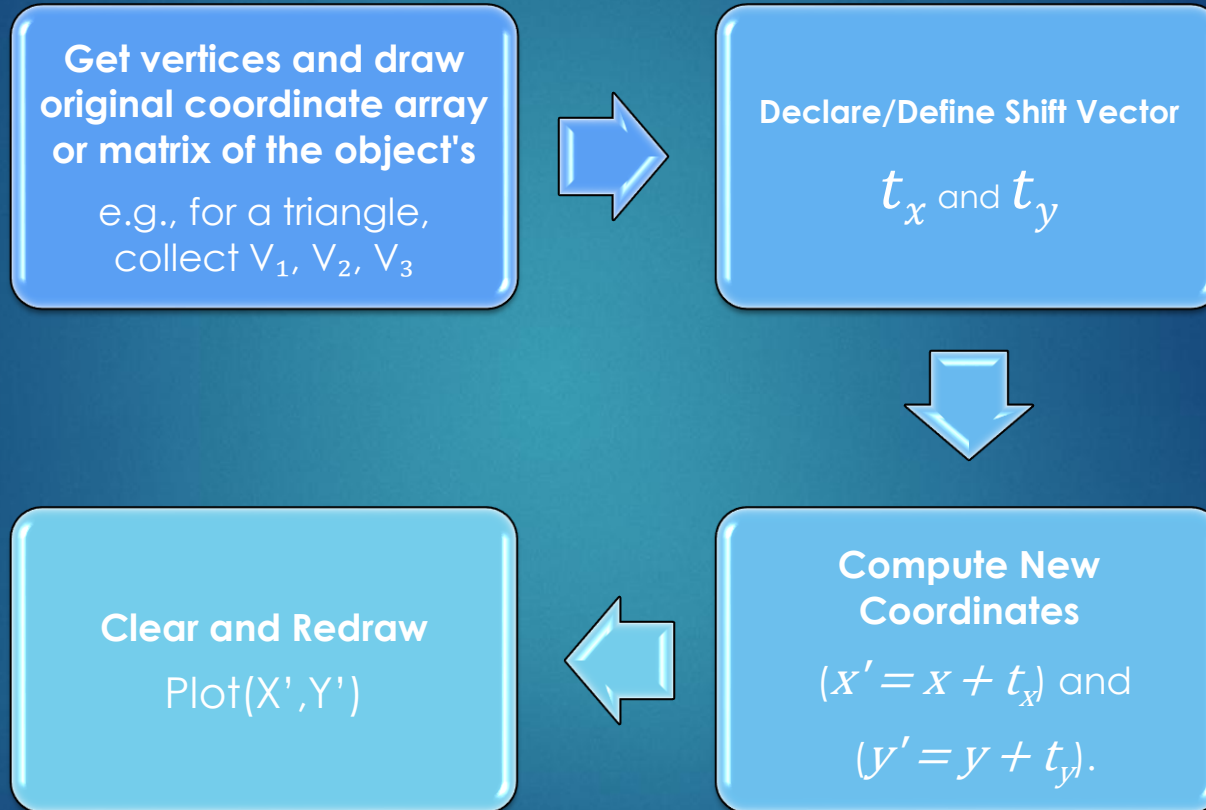
$$C = \begin{bmatrix} 6 \\ 4 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$$

$$C = (8, 6)$$



2D TRANSLATION

ALGORITHM STEPS





**MORE PRACTICE MAKES
YOU PERFECT**





Always Seek Knowledge



THE END

